# Portfolio app: system reference

**What this is.** A complete technical description of the personal portfolio application built on top of the open-source TradingAgents project, as of 2026-10-24 (branch `portfolio-app`). It covers what the app does, every model, what data is used and saved, how forecasts are made and evaluated, every metric, the measured results, and the known limits. It is written to be reviewed: by the owner, and by other AI systems asked for critique (see `REVIEW_PROMPT.md`).

**The account.** One Robinhood **Roth IRA** (a cash account: purchases need settled cash; sales settle T+1; no margin, no short selling; options at most level 2 if approved). About **$5,844**, kept as two bookkeeping sleeves inside the one IRA: **Investing** (long-term, about $4,844) and **Trading** ($1,000 assigned initially, then changing only with its own profit and loss and explicit transfers). The app plans and records; **it never places orders**. The owner trades by hand.

**Ground rules the whole system follows.**

1. No look-ahead: every feature uses only information available at its date.
2. Every experiment is recorded, whatever it found, so multiple testing is visible.
3. Paid AI is opt-in per action, cached, budgeted and logged; numbers never come from a language model, only from fitted models or arithmetic.
4. Records are append-only: corrections are new records pointing at old ones.
5. Data is downloaded once and then only what is new.

## 1. Architecture

```
TradingAgents (upstream)   multi-agent LLM analysts (market, news, fundamentals, sentiment)
        |
tradingagents/dataflows    vendors: Yahoo Finance (prices, profiles, statements, analysts, news),
        |                  SEC EDGAR (company facts, filings), FRED/ALFRED (economic series),
        |                  local market store (SQLite + pickles): incremental cache
        |
tradingagents/optimizer    ETF/stock portfolio optimizer, risk levels, plan, daily orders (T+1),
        |                  auto-invest, paper tracker, backtest, market overview
tradingagents/research     ETF return forecasting: features, models, walk-forward, risk models,
        |                  neural (LSTM) and pretrained (Chronos-2) challengers, ticker research
        |                  (moat, quality checklist, dividends/splits), compute (CPU/GPU)
tradingagents/stocks       stock-only research: point-in-time fundamentals and macro, event targets,
        |                  batch forecasts, news sentiment, recommendations, outcomes
tradingagents/ledger       SQLite books: sleeves, events, settlement, recommendations, decisions,
        |                  AI gateway (budget/cache), forecast and stock-run records, migrations
tradingagents/webapp       local web app (127.0.0.1:8765), stdlib HTTP server + static pages
cli/main.py                command line: web, suggest, track, book ..., stocks ...
experiments/               studies that produce the evidence (walk-forward, ablations, risk)
```

Hardware: Windows 11 laptop, i5-11400H, 16 GB RAM, NVIDIA RTX 3050 Ti (4 GB). Python 3.13, XGBoost 3.4.1 (CUDA build), scikit-learn 1.9.1, PyTorch 2.11 (CUDA 12.8), arch 8.0, transformers 5.20. Everything numerical runs locally.

## 2. Data sources and what is saved

| Source | What | How it is cached | Point-in-time handling |
|---|---|---|---|
| Yahoo Finance (yfinance 1.7) | Daily OHLCV (auto-adjusted for splits and dividends), quote profiles, financial statements (4 annual + quarterly), analyst recommendations, price targets, EPS/revenue estimates, upgrades/downgrades, earnings dates, dividends, splits, shares outstanding, news | Local market store: bars as one pickle per ticker, only new sessions fetched; snapshots in SQLite with TTLs (news 3 h; statements, analyst histories, actions 7 days; profiles daily) | Prices are as-of. Statements and analyst lists are **today's** versions (restatements and deletions invisible), so they are used only for display, not for model training, except where noted |

| Source | What | How it is cached | Point-in-time handling |
|---|---|---|---|
| SEC EDGAR company facts (XBRL) | Revenue, gross profit, cost of revenue, operating and net income, operating cash flow, capex, assets, liabilities, equity, shares outstanding, with **filing dates** | Compact per-stock JSON (`sec_pit2/`), refetched weekly; rate-limited (8/s) with retries | A fact counts from the **day after** its filing date; restatements are new facts with later filing dates and never rewrite earlier history |
| SEC EDGAR filings | Latest 10-K business section (Item 1, incl. Competition) | Saved per accession number, fetched once per filing | Used for the AI moat reading only |
| FRED / ALFRED | CPI, unemployment, payrolls, NFCI (all vintages); 10y-2y spread, Baa-10y spread, fed funds (unrevised) | JSON per series, refreshed weekly | Revised series: each observation's latest vintage published **before** the date, counted from the day after publication. Unrevised: usable the day after the observation |
| S&P 500 member list | Ticker, name, GICS sector, CIK, date added | CSV, refreshed weekly | Today's members only (survivorship), but each member enters training only from its date added |
| News (Yahoo) | Headlines, publisher, link, publication time | `stock_news` table, append-only, with first-seen time | Collected prospectively; no historical news exists in the system |

**Saved records (SQLite `portfolio.sqlite3`, migrations 1-4, append-only by triggers):** accounts, events (contributions, fills, transfers, adjustments, voids), settlement and reservations, settings and their history, recommendations (official plans, with code version, settings and prices used), decisions, orders, AI usage (tokens, latency, cost), AI result cache, research_trials (every experiment), forecast_runs/forecasts (ETF and stock forecast runs with model file hashes), stock_runs, stock_forecasts, stock_news, stock_ideas, stock_idea_decisions, stock_outcomes.

Other files: `paper_tracker.json` (the Day-1 paper record), model artifacts (pickles with SHA-256 in the run record), `results/` (saved analysis reports, market briefs, moat readings), `experiments/output/` (study predictions and summaries).

## 3. The portfolio optimizer (Investing and Trading sleeves)

**Universe.** About 40 long-lived ETFs (US total market, sectors, international, bonds, gold, commodities, leveraged), the S&P 500 (cached member list), and disruptive-theme stocks and small-cap funds: about 566 tickers, plus holdings. Each is scored cheaply (expected growth above cash per unit of crash risk); the best 60 plus holdings go to the optimizer.

**Risk dial (1-10).** A stress-loss limit on the whole portfolio:

| Level | 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 | 9 | 10 |
|---|---|---|---|---|---|---|---|---|---|---|
| Limit | 1% | 2.5% | 5% | 10% | 15% | 20% | 30% | 40% | 50% | none |

**Stress loss**, two models:

- *Worst case* (default): sum of weight x each fund's own worst historical peak-to-trough drop (young funds: a long-lived stand-in, or a 50% floor under 10 years of history).
- *Crash scenarios* (optional): each past crash in SPY, TLT, GLD or DBC replayed on the mix; absent funds estimated by downside beta.

**Solver.** Maximize expected growth minus a risk penalty, subject to sum(weight x stress loss) <= limit, per-asset caps (35% of the portfolio; at most 25% of the loss budget per asset), long only. Augmented Lagrangian with accelerated projected gradient (FISTA). Optional **growth target**: bisection for the least risk that reaches a target growth, never past the dial's cap.

**Expected returns.** The standard rule (vol_prior): 0.3 Sharpe x volatility, less half the variance, shrunk for short histories. Optionally a saved forecast model's output (relative mode: only differences between assets move money, scaled by the model's measured within-date calibration, clipped at 2 sigma). The default is the standard rule.

**Orders.** The next session's orders under T+1: buys use settled cash only; sales of the cash fund (SGOV) fund the following session's buys; trades under 0.5% of the account (or $1) are skipped. No day trading. **Auto-invest** splits a daily budget into Robinhood recurring buys (every market day, week, 2 weeks or month; $1 minimum assumed).

**Records.** Official recommendations saved per session (before 3:45 pm New York for that day, else the next session), with code version, settings and prices; decisions and fills recorded separately. The **paper tracker** follows two portfolios from Day 1 ("followed the plan" vs "kept as is") under the same T+1 rules.

**Measured (walk-forward 2008-2026, ETFs, monthly rebalances).** The worst-case model never broke its limit in 274 one-year holds; the crash-scenario model broke it by up to 1.5x. Growth estimates ran 0.3-1.3 points a year above what was realized. The optimizer's Sharpe (0.52-0.60) is close to 60/40's (0.55): it controls drawdown, it does not add return. Every setting trailed holding VTI or SPY over 2012-2026.

## 4. ETF return forecasting (`tradingagents/research`)

**Target.** Log return **above cash** (BIL) over 5, 20 or 60 trading sessions, from the decision date's close.

**Features (22), each from data up to the decision date's close, ranked across assets within each date into [-0.5, 0.5]:** momentum 1/3/6/12-1 months, 1-week reversal, volatility 1m/3m, downside volatility, beta to SPY (1y), distance from the 52-week high and the 200-day average, 1-year drawdown, high-low range, dollar volume, its trend, Amihud illiquidity. Market levels (standardized on training rows): VIX, VIX/VIX3M slope, 10-year yield's 3-month change, 13-week bill rate, SPY 3-month return and 1-month volatility.

**Models.** zero, train_mean, vol_prior (the incumbent rule), momentum (12-1 rank), Elastic Net (penalty chosen on the last fifth of training dates), XGBoost (pseudo-Huber), XGBoost ranking (pairwise within date, mapped to returns by a line on held-back data), XGBoost quantiles (5/50/95). All wrapped by `Probabilistic`: quantiles from standardized held-back residuals (or direct quantiles), P(gain), and a bootstrap "model uncertainty" (spread of the mean over 5 refits on resampled data). Challengers: a small LSTM (about 10k weights, 60 daily returns + features) and **Chronos-2** (Amazon, 120M parameters, Apache-2.0, zero-shot on the last 1,024 log prices; contamination risk unknown).

**Walk-forward.** Yearly refits from 2012; training rows only where the label **ended** before the test block (purge, embargo = horizon); shared folds for all assets; all uncertainty computed over dates (moving-block bootstrap, 12-date blocks); Holm-adjusted p-values across the models in a trial.

**Results (ETF universe, 2012-2026).**

| Horizon | Model | IC (95% CI) | Within-date calibration | 5-95% coverage |
|---|---|---|---|---|
| 20 | standard rule | 0.067 (0.023, 0.110) | -0.70 | 89% |
| 20 | Elastic Net | 0.111 (0.007, 0.219) | 1.01 | 89% |
| 20 | XGBoost | 0.113 (0.043, 0.192) | -0.28 | 88% |
| 20 | LSTM | 0.114 (0.042, 0.182) | 1.11 | 89% |
| 20 | Chronos-2* | 0.101 (0.047, 0.158) | -0.06 | 87% |
| 60 | standard rule | 0.118 (0.045, 0.193) | -0.89 | 90% |
| 60 | XGBoost | 0.212 (0.111, 0.326) | 0.92 | 88% |
| 60 | XGBoost ranking | 0.220 (0.129, 0.317) | 0.03 | 83% |
| 60 | Chronos-2* | 0.207 (0.127, 0.306) | 0.33 | 90% |
| 5 | XGBoost quantiles | 0.092 (0.065, 0.124) | 0.75 | 87% |

Findings: learned models rank ETFs better than chance and better than the standard rule; **no model times the market** (date-average forecast vs outcome correlation about 0 or negative); P(gain) is worse than the base rate; in the portfolio test after 10 bp costs **no forecast beat the standard rule** (the best roughly tied with 5x the trading).

**Chronos-2 stacking ("+chronos").** Chronos-2's median forecast and 5-95 width as two extra inputs. Before its release: XGBoost ranking IC 0.220 -> 0.236 at 60 sessions (top fifth net +1.61% -> +2.15% per period); plain XGBoost slightly worse at 20. After its release (10-12 dates, the only uncontaminated test) adding it made every 60-session model slightly worse. It is an option, not a default.

**Live record.** Every saved run is scored once its horizon has passed (IC per run date, top fifth vs all). Chronos-2's forecasts are recorded at each new close when the app starts: the clean test of a model that may have seen past prices.

**Combined forecast.** Each model's latest forecasts at a horizon are shown under each ticker with an evidence-weighted combination: weight = walk-forward IC where Holm p < 0.05; untested runs, runs that give every ticker the same forecast, and Chronos-2 get weight 0.

**Stock universe for the ticker page (S&P 500, price features only, survivorship-biased).** 60 sessions: XGBoost ranking IC 0.066 (0.038, 0.093), top fifth +1.28%/period net, Holm p < 0.001. 20 sessions: nothing significant; trained on all data, Elastic Net and XGBoost regressors gave every stock the same forecast (only market-wide splits survived). Stock pages use a stock model at 60 sessions only.

**Risk models.** Volatility (QLIKE, lower is better): GARCH(1,1)-t best at every horizon (20 sessions: 0.328 vs EWMA 0.391 vs 10-year sample 0.698). Covariance: the 10-year sample did best for the optimizer (Ledoit-Wolf, factor and EWMA on 1 year underestimated next month's variance by 30-40%).

## 5. Ticker research (ticker page, any symbol)

- **Wall Street**: consensus (strong buy to strong sell counts over 4 months), price targets, EPS and revenue estimates with revisions, upgrades/downgrades history, earnings estimates vs reported. From Yahoo; today's versions.
- **Quality checklist**:
- *Economic moat (numbers)*: pricing power (gross margin level, floor and stability against sector-specific bars), ROIC every year (operating income x (1 - own tax rate) / average invested capital, excess cash beyond 2% of revenue deducted), operating margin, positive FCF every year, steady revenue growth; banks/insurers get ROE, ROA, net margin, book value per share growth. Verdict: strong (4-5 of 5), some (2-3), few.
- *Moat sources (AI, paid, optional)*: the latest 10-K's business section read by the configured model; switching costs, network effects, cost advantage, intangibles, efficient scale, each rated with a **verbatim quote verified against the filing** (unverified quotes downgrade the rating to "unclear"); durability and named threats.
- *Financial strength*: debt/equity (flag > 2), interest coverage (flag < 3), current ratio (flag < 1.0), a high-rate downturn test (operating income -30%, all debt refinanced +2 points; flag < 1.5x, thin < 3x).
- *Profitability and cash flow*: FCF (TTM, margin, conversion), ROIC vs 15%, gross / operating / net margin trends (expanding / stable within 1 point / shrinking).
- *Valuation*: forward P/E vs the company's own year-end P/Es (prices restored from dividend-adjusted closes), EV/EBITDA, FCF yield, and a **reverse DCF**: the 10-year FCF growth today's EV implies at a 9% discount rate and 2.5% terminal growth, judged against the revenue record and analysts' next-year estimate.
- **Dividends, splits, share count**: dividend history (split-adjusted), TTM yield, years of raises (by typical payment, not calendar totals), 5-year growth, cuts and suspensions; splits; share count restated for splits, with dilution (>= +2%/yr) and buybacks (<= -2%/yr); funds excluded from share-count analysis.
- **Forecasts**: the standard rule's yearly figure and each horizon's model forecast, flagged when the model never saw this kind of asset.
- **News**: chronological headlines.

# 6. Stock research module (`tradingagents/stocks`)

## 6.1 Scope and inputs

Stocks only. Tickers from a pasted list, a CSV (ticker/symbol column), the saved watchlist, and the account's holdings. Each is classified: S&P 500 members are stocks; others must be Yahoo `quoteType == EQUITY`. Funds, mutual funds, crypto, futures, options, indexes, currencies and unknown symbols are turned away with the reason; one failed lookup never stops the rest. Market, sector and economic series remain contextual inputs.

## 6.2 Targets (defined before training; thresholds frozen and versioned)

Decision after the close of date t. sigma_h = daily EWMA volatility at t (lambda 0.94) x sqrt(h).

- **Tradable return**: enter at the **next** close (t+1), exit h sessions later; log, above cash (BIL) and separately above SPY. An after-close forecast never trades at that close.
- **Terminal return** (path events): log(close[t+h] / close[t]).
- **Drawdown from entry**: min(low[t+1..t+h]) / close[t] - 1 (distinct from terminal loss).
- **Large loss**: terminal return <= -1.5 sigma_h. **Deep drawdown**: drawdown <= -(exp(1.5 sigma_h) - 1).
- **Breakouts**: barriers = highest high and lowest low of the 20 sessions through t. Up breakout = any high in t+1..t+h above the upper barrier; down = any low below the lower. Both may occur; a single bar crossing both is recorded (`both_same_bar`) without inventing an order. **Failed breakout**: crossed, but closed the period back inside.
- **Trend state** (exactly one of four): *up*: r >= 0.5 sigma_h and path efficiency (|r| / sum |daily log returns|) >= 0.30; *down*: mirror; *sideways*: |r| <= 0.35 sigma_h and the whole high-low range <= 1.25 sigma_h (so a volatile round trip is not sideways); *unclear*: everything else. Events overlap trend states freely.
- **Current trend** (detection, not forecast): price above a rising 50-day average above the 200-day (+1), mirror (-1), else 0. A feature and a display only.
- Threshold version at the time of writing: `thr-14adb1ddde`.

Observed rates (20 sessions, 2006-2026, S&P 500): up breakout 58%, down 41%, large loss 6%, deep drawdown 12%; trend mix: unclear 52%, up 20%, sideways 18%, down 10%.

## 6.3 Features (four groups, used for ablation)

- **technical** (26): the ETF price features plus 20- and 50-day average slopes, position in the 20-day and 52-week ranges, strength vs SPY and vs the stock's sector ETF over 1 and 3 months, 1-week volume vs 3 months, current trend.
- **market** (6): as in section 4.
- **fundamental** (14): revenue growth (annual and latest quarter vs a year earlier), gross, operating, net and FCF margins, ROA, liabilities/assets, dilution over a year (split-adjusted), earnings/sales/FCF/book yields on the market value at the date (raw price, undoing later dividends and splits, x shares then reported), filing age. From SEC facts as known at the date.
- **macro** (8): CPI inflation (12 months), 3-month change in unemployment, payroll growth, NFCI, 10y-2y, Baa-10y spread and its 3-month change, fed funds. As first published.

Cross-sectional features are ranked within each date; market and macro stay levels (standardized on training rows). Missing values count as 0 (the middle). XBRL facts start 2009-2011, so fundamentals are missing for early years (coverage about 70-80% overall). **Estimated trading cost**: Abdi-Ranaldo spread from daily high/low/close over the 21 sessions before the date, plus 2 bp slippage a side; shown separately, subtracted only in "net" figures. Analyst histories are **not** used in stock training (Yahoo's lists are today's versions; not defensibly point-in-time). News is **not** used in training (no history).

## 6.4 Models

Pooled across stocks (one model per target and horizon, not per stock):

- Returns: momentum (benchmark), Elastic Net, XGBoost quantiles (each via `Probabilistic`); XGBoost regression for the return above SPY.
- Events (up/down breakout, large loss, deep drawdown): base rate (null), L2 logistic (penalty by validation log loss), XGBoost logistic (eta 0.05, early stopping).
- Trend state: multinomial logistic and XGBoost softprob.
- Severity: XGBoost quantiles on the drawdown from entry (5th, 50th percentiles).
- Volatility: EWMA vs GJR-GARCH(1,1) with Student-t errors (study only).
- Training universe: S&P 500, each stock from its date added; models are saved per (cutoff, horizon, features, thresholds, code version) and reused for later batches that day.

## 6.5 Batch pipeline

Inputs checked -> shared data fetched once (bars incremental; SEC facts and Yahoo actions 4 at a time, rate-limited, retried, cached) -> per horizon: panel, models fitted or loaded, every S&P 500 stock forecast at the cutoff (so requested stocks get a percentile) -> news collected and summarized -> evidence attached from the latest study -> saved. A stock with stale or missing prices, no trading volume, or under a year of history **abstains** with the reason; failures are isolated per ticker. Per-ticker progress is reported. Measured: a first batch of 5 tickers at one horizon took 202 s including training; a second batch the same day reuses the models.

Each forecast (per stock and horizon) saves: expected return above cash (evidence-weighted combination of Elastic Net and XGBoost quantiles), median, 5/95%, P(gain), each model's mean and their spread, return above SPY, estimated round-trip cost, net expected return and its percentile among the S&P 500, trend probabilities (up/down/sideways/unclear), breakout probabilities with the barrier levels, large-loss and deep-drawdown probabilities, drawdown 5th/50th percentiles, current trend, filing age, the training base rates, the evidence status per target (backed / weak / none / untested), news summary, reasons, and source.

## 6.6 News

Collected per stock with publication and first-seen times; identical headlines are one record and near-duplicates (80% shared words within 3 days) are linked. *About* (named in the first 40 characters) vs *mentions*. *Fact* vs *opinion* vs *unclear* by rule. Event tags by rule: earnings, guidance, financing, buybacks/dividends, litigation, regulation, management, M&A, analyst actions, products/contracts. Sentiment = P(positive) - P(negative) from **FinBERT (ProsusAI/finbert) run locally** (fallback: a Loughran-McDonald word list, named in the record); averaged over headlines *about* the stock in 7 days. No headline means "sentiment unknown", never neutral. Optional paid reading for a shortlist (default 5) through the AI gateway: structured events with facts separated from interpretation; citations must point at real headline numbers.

## 6.7 Recommendations (reviewable, never executed)

For the Trading sleeve by default, from a run at 20 sessions: **buy candidate** when net expected return > 0, return evidence *backed* by the final test, P(large loss) <= 15%, and no data flag; at most 5 names; equal dollars, each <= 25% of the sleeve, together within **settled** cash; fractional shares; orders under $1 dropped. **Watch** when positive but not backed or flagged. **Avoid / consider reducing** when negative: **never a short order**; a sale is at most what is held. Earliest execution: the session after the cutoff; hold through the horizon (no day trades). Before sizing, the whole IRA (stocks and ETFs) is priced and each candidate's correlation and beta to it, and the IRA's beta to SPY after adding it, are shown (funds are not looked through to sectors). Decisions are separate records; fills stay in the ledger.

## 6.8 Three evaluation systems, kept separate

1. **Historical strategy simulation**: the walk-forward study (section 7), using only information available at each decision date.
2. **Prospective Day-1 record**: each saved recommendation's buys vs putting the same dollars into SPY on the same dates (frozen static policy), at the same execution assumptions, once matured. Old recommendations are never regenerated.

3 **Actual performance**: the ledger's fills and values.

Matured forecasts get their realized outcomes in `stock_outcomes`, scored in a live scorecard (IC per run date, 5-95% coverage, Brier vs base rate per event).

### 6.9 Next steps, justified by the measured evidence

1 **Test the ranking objective (XGBoost rank within date) and cross-sectionally ranked labels** under the point-in-time design: the market component dominates stock returns, so regressors collapse to market-wide splits; a within-date ranking target removes it. The earlier, looser S&P 500 study found IC 0.066 at 60 sessions with it.

2 **Point-in-time index membership** (members later removed) to remove survivorship bias.

3 **Use GJR-GARCH-t volatility** for sigma_h and the forecast scale (it beat EWMA).

4 **Keep breakout probabilities as risk context**; test whether they add anything beyond distance-to-barrier and volatility (a two-feature logistic baseline).

5 **Do not add a sequence model (PatchTST, VSN+LSTM) or a pretrained model yet**: no return model has passed the final test, the target is dominated by noise, and on ETFs the LSTM and Chronos-2 did not beat boosted trees. Revisit when a simple model shows confirmed skill.

6 **Fix tree counts** (no early stopping in deployment, or a fixed number from the study) if identical CPU/GPU results are wanted.

## 7. Validation methods and every metric

**Folds.** Chronological, shared by all assets; training uses only rows whose labels ended before the test block (purge, embargo = horizon); preprocessing (standardization, penalty choice, early stopping) fitted inside each training fold on its last fifth of dates (also purged). Stock study: blocks of 2 years from 2012; **development 2012 to June 2024**, where models are compared, the best chosen and ensemble weights learned; **final test July 2024 on**, scored once with everything frozen.

**Units of uncertainty.** Dates, not rows: hundreds of correlated stocks on one date are one market state. Intervals by moving-block bootstrap over dates.

| Metric | Definition | Good sign |
|---|---|---|
| IC | Spearman rank correlation of forecast and outcome across assets on each date, averaged | > 0, interval above 0 |
| p (Holm) | Two-sided p of the IC t-stat, Holm-adjusted across the models in a trial | < 0.05 |
| $R^2$ out of sample | 1 - SSE(model) / SSE(zero forecast) | > 0 |
| Calibration (pooled / within date) | Slope of outcome on forecast, all rows / within each date | about 1 |
| Timing correlation | Date-average forecast vs date-average outcome | > 0 (it is about 0) |
| Pinball loss, coverage | Quantile loss at 5/50/95%; share of outcomes inside 5-95% | coverage about 90% |
| Brier, log loss | For P(gain) and every event, against the training base rate (null) | below the null |
| Brier skill | 1 - Brier / Brier(null), with a date-block bootstrap interval | > 0, interval above 0 |
| AUC | Ranking of events by probability | > 0.5 |
| Reliability table | Forecast probability tenths vs observed frequency | on the diagonal |

| Metric | Definition | Good sign |
|---|---|---|
| Alerts | Top tenth by probability on each date: precision, recall, missed, false alarms, lift | lift > 1 |
| Failed breakouts | Share of alerted, realized breakouts that closed back inside | low |
| Crash lead time | Sessions from the forecast to the first breach of the drawdown threshold | longer |
| Trend scores | Multinomial log loss vs the training mix; accuracy vs always guessing the commonest state; recall per state | above the nulls |
| Severity | Share of drawdowns below the forecast 5th percentile (target 5%); pinball loss | about 5% |
| Top fifth vs all | Per date, equal-weight top fifth by forecast minus all, gross and net of costs on turnover | > 0 |
| Portfolio | Long-only top fifth by net expected return, non-overlapping holds, vs all stocks, momentum, trend rule: mean, annualized, total, max drawdown, turnover, by period | above benchmarks |
| QLIKE | realized/forecast variance - log(ratio) - 1 | lower |
| Ablation | IC of each feature-group set on development | added groups help |

Every run (ETF, stacking, stock, ablation variant) is stored in `research_trials`.

## 8. Stock study results (2026-10-24, trials `stock_study:h5/h20/h60`, rescored)

Universe: today's S&P 500 (503 stocks), each from its date added; all four feature groups; thresholds `thr-14adb1ddde`. Development 2012 to June 2024 (314 dates at 5 sessions, 157 at 20 and 60); **final test July 2024 to October 2026 (59 dates at 5 sessions, 30 at 20 and 60)**. Intervals: 95% moving-block bootstrap over dates. "Ranked" = dates on which the model told stocks apart at all (a model that gives every stock the same number has no rank that day).

**Returns (IC, development / final test).**

| Horizon | Momentum | Elastic Net | XGBoost quantiles | Equal ensemble |
|---|---|---|---|---|
| 5 | 0.015 (-0.005, 0.036) / **0.042 (0.017, 0.089)**, Holm p 0.57 | 0.011, ranked 15 of 314 / -0.015, ranked 2 of 59 | 0.011 (-0.031, 0.046), ranked 169 / -0.034, ranked 9 | 0.011 (-0.009, 0.031) / 0.040 (0.012, 0.088) |
| 20 | -0.019 (-0.044, 0.011) / -0.015 (-0.035, 0.024) | 0.003 (-0.021, 0.022) / -0.007 (-0.042, 0.015) | 0.024 (-0.019, 0.064), ranked 96 of 157 / ranked 0 of 30 | 0.003 / -0.007 |
| 60 | 0.016 (-0.018, 0.068) / -0.034 (-0.054, 0.006) | 0.014 (-0.013, 0.036) / 0.002 (-0.051, 0.028) | 0.022 (-0.012, 0.063), ranked 104 / ranked 0 of 30 | 0.016 / -0.010 |

Return above SPY (relative model): IC 0.001 to 0.025 in development, -0.036 to 0.001 in the final test. **No stock return model has skill that survives the final test and the multiple-testing correction.** The tree models often collapse to market-wide splits only (the same forecast for every stock on a date): early stopping on a noisy target keeps very few trees. No development-period model was significant, so the ensemble weights are equal.

**Events (ensemble of logistic and boosted trees; Brier skill vs the base rate, AUC, lift of the top tenth).**

| Horizon | Up breakout | Down breakout | Large loss | Deep drawdown |
|---|---|---|---|---|
| 5 dev / test | +0.300 (0.263, 0.333), AUC 0.82 / **+0.281 (0.234, 0.338)**, AUC 0.82, lift 2.6 | +0.254 / **+0.200 (0.132, 0.338)**, AUC 0.82, lift 2.9 | -0.047 / +0.003 (-0.016, 0.021) | +0.005 / +0.026 (-0.027, 0.107) |
| 20 dev / test | +0.230 (0.197, 0.254) / **+0.230 (0.218, 0.287)**, AUC 0.77 | +0.183 / **+0.197 (0.186, 0.278)**, AUC 0.76 | -0.103 / -0.006 (-0.009, 0.003) | -0.050 / +0.019 (0.009, 0.029) |
| 60 dev / test | +0.143 / **+0.190 (0.144, 0.207)**, AUC 0.75 | +0.114 / **+0.107 (0.097, 0.146)**, AUC 0.70 | -0.083 / -0.002 | +0.007 / -0.081 (-0.156, 0.030) |

Base rates (final test): up breakout 30% / 52% / 71% at 5 / 20 / 60 sessions; down 22% / 43% / 58%; large loss 5-7%; deep drawdown 9-13%. Of alerted breakouts that happened, 43-55% failed (closed back inside the range). Deep-drawdown alerts led the first breach by about 3, 10 and 31 sessions on average, but their skill is not reliable (development negative). **Breakout probabilities are the only stock forecasts with confirmed skill**; they are probably largely mechanical (distance to the barrier relative to volatility), so they describe risk, not return.

**Trend states**: no skill. The boosted trees predict "unclear" (the commonest state) almost always; multinomial logistic is worse than the base mix in development.

**Severity** (drawdown 5th percentile): 4.6-8.4% of outcomes fell below it (target 5%): roughly calibrated, slightly too narrow at 5 sessions in the final test.

**Portfolio** (long-only top fifth by net expected return, non-overlapping holds, per period, above cash): 20 sessions development +1.41% vs all stocks +1.17% (difference +0.24%, interval -0.02% to +0.51%), final test +0.07% vs +0.80% (difference -0.73%); momentum was best in the final test. 60 and 5 sessions: no difference. Turnover about 0.5-0.7 per period. **No robust portfolio edge.**

**Ablation** (20 sessions, development): Elastic Net IC about 0 with any feature set; boosted trees 0.004 (technical only), 0.055 (+ market), 0.104 (+ fundamental), 0.058 (0.003, 0.083) with all four groups. The tree figures are computed only on dates on which the model ranked anything, and their intervals are mostly undefined: **not reliable evidence that any group adds value**.

**Volatility** (20 sessions, 40 stocks, 6,720 forecasts, QLIKE, lower is better): GJR-GARCH(1,1) with Student-t errors 0.351, EWMA 0.408. GJR-GARCH-t is better.

**Compute** (measured): panel build about 28-30 s per horizon (503 stocks, 2006-2026); a first batch of 5 tickers at one horizon took 202 s with training, and reuses its models thereafter. Boosted-tree event model on 181k rows: CPU 3.0 s, GPU 1.5 s; on 90k rows: CPU 1.8 s, GPU 2.4 s (the GPU is used automatically only from 200k rows). CPU and GPU fits are **not numerically identical**: early stopping chose different numbers of trees (53 vs 58; 165 vs 245), giving probability differences of 0.01-0.03 on average and up to 0.13. Runs record the device used.

**What the app does with this.** A stock forecast's evidence status per target comes from the final test: returns are *none* at every horizon today, breakouts are *backed*. Recommendations therefore list every stock as *watch*, never as a buy candidate, until a return model passes the final test. The forecasts and probabilities are still saved and shown, labelled.

**The earlier S&P 500 study** (price features only, today's members with no entry dates, yearly refits, XGBoost ranking objective) found IC 0.066 (0.038, 0.093) at 60 sessions. The point-in-time study here did not include the ranking objective; testing it under the stricter design is the first next experiment.

**Scoring fix.** The first scoring of this study stored the Brier-skill interval high-first; it was rescored from the saved predictions (no refitting) and recorded as new trials.

## 9. Compute, costs and controls

- `cpu`, `gpu` or `auto` (GPU only for boosted-tree jobs of 200,000+ rows, and for Chronos-2 and the LSTM); one GPU job at a time (lock file); a GPU failure reruns on the CPU; never a cloud fallback. Measured: XGBoost 200k x 40, 300 rounds: CPU 2.4 s, GPU 1.6 s; Chronos-2, 64 series: CPU 4.8 s, GPU 1.0 s, medians agree to 6e-8.

- AI gateway: cache key = task + provider + model + schema version + information cutoff + input hash; concurrent identical requests wait for one; daily/monthly dollar and request budgets (defaults $1/day, $10/month); **a model with no configured price cannot be called**; `TRADINGAGENTS_NO_PAID_AI=1` refuses all; every request's tokens and cost logged. Page loads, polling, replays and tests never pay. Currently no price is configured, so paid features (AI ratings, market brief, moat sources, news reading) are blocked until one is set.
- Data: prices, statements, filings, economic data and news are free; downloads are incremental and cached (section 2).

## 10. How to run

```
.venv\Scripts\python.exe -m cli.main web                # the web app, http://127.0.0.1:8765
.venv\Scripts\python.exe -m cli.main suggest            # today's plan and orders
.venv\Scripts\python.exe -m cli.main stocks run AAPL MSFT --horizons 5,20,60
.venv\Scripts\python.exe -m cli.main stocks run --watchlist --holdings --csv list.csv
.venv\Scripts\python.exe -m cli.main stocks show [--run ID] [--horizon 20]
.venv\Scripts\python.exe -m cli.main stocks ideas --horizon 20
.venv\Scripts\python.exe -m cli.main stocks decide IDEA TICKER accepted
.venv\Scripts\python.exe -m cli.main stocks outcomes     # record matured outcomes; scorecard; Day-1 record
PYTHONPATH=. .venv\Scripts\python.exe experiments\stock_study.py 5 20 60
```

API (no stock page yet): `POST /api/stocks/run`, `GET /api/stocks[?run=]`, `POST /api/stocks/ideas`, `POST /api/stocks/decide`, `POST /api/stocks/watchlist`, `GET /api/stocks/outcomes`, `POST /api/stocks/news-ai` (paid).

## 11. Known limits

- **Survivorship**: the stock universe is today's S&P 500. Entering each stock from its date added removes the pre-inclusion run-up, but stocks later removed are missing.
- **One path of history**, mostly a US bull market (2012-2026).
- **No intraday data**: breakout ordering within a bar is unknown (recorded, not invented); execution is modelled at the next close, not the open.
- **Costs** are estimated from daily bars (Abdi-Ranaldo); market impact is ignored (order sizes are tens to hundreds of dollars).
- **Fundamentals** begin 2009-2011 (XBRL); some tags are missing for some companies.
- **Analyst data** (Yahoo) is today's version and is excluded from stock training.
- **News** exists only from when collection began; no news backtest is possible yet.
- **Chronos-2** may have seen the test prices in pretraining; only the live record is clean.
- **Paid AI** features are blocked until a model price is configured.
- **The app never trades**; recommendations are research for the owner to review.

## 12. Daily stock decisions, capital settings and challengers (added later on 2026-10-24)

Full detail: `docs/DECISIONS.md`. In brief:

- **Capital settings** (Settings > Capital; `book capital`; `/api/capital`): Trading capital $1,000 (net funding target, not a daily budget), Investing capital $4,800, daily investment $27.61 (planned purchase per session, bounded by settled cash). Validated, versioned, snapshotted into every record; changes produce reviewable transfer proposals and never move money or retrain models.

- **Execution contract** and one **position engine** for labels, simulation and the shadow account: next-close entry, close-confirmed levels filled at the next close, exact deadlines, no same-day exits, T+1, dividends as cash, executable prices.
- **Foundations fixed** after verification: point-in-time membership (fja05680/sp500; survivorship partial), members-only ranking with requested-ticker invariance, unknown capex, split-consistent valuation, expm1 barriers, no zero cash returns, per-series FRED lags, as-of pricing, missing risk blocks approval, limits including holdings, evidence bound to the exact design. The original stock study's evidence no longer applies (new thresholds version `thr-016cb9da59`).
- **Decisions**: ENTER / PAPER ENTER / WAIT / HOLD / REDUCE / EXIT / REVIEW with size, execution date, expected net P&L and spread, P(profit), P(large loss), levels, deadline, reasons and blocks.
- **Decision study**: XGBoost learning-to-rank IC 0.020 (0.005, 0.036) in development; trade-quality probabilities worse than the base rate; the complete policy trails cash-flow-matched SPY (best development policy +210% vs SPY +358% over 2013 to mid-2024; exploratory +5.0% vs SPY +43% and cash +9.9%). **Not validated**: the app is in experimental mode (WAIT; PAPER ENTER feeds the shadow account).
- **Challengers** on 150 stocks: no ranking skill for PatchTST, Chronos-2, TimesFM 2.5; Kronos-mini 0.041 (0.016, 0.067) in development but likely contaminated and 0.014 (not significant) after.